Lecture Notes

Lecture 11

Dual Spaces, Functionals, and Multilinear Forms

We have studied vectors and the operators that move them. This lecture studies the *measurements* one can make on vectors: the linear functionals $\varphi : V \to \mathbb{F}$, which assemble into the *dual space* V^*. In an inner product space the Riesz theorem of Lecture 8 already tied functionals to vectors—a bra $\langle u|$ is a dual vector. Here we build the dual abstractly, without an inner product: dual bases, the dual map (the transpose, freed from coordinates), bilinear and multilinear forms, and the covariant/contravariant bookkeeping with Einstein's summation convention that runs through tensor calculus and relativity. This is Part II of the course: the multilinear algebra underneath tensors.

Remark 11.1 (Standing scope for Lectures 11–12). The core is finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$. Here $V^* = \mathcal{L}(V, \mathbb{F})$ is the algebraic dual. In a complex inner-product space the physics-convention Riesz map $V \to V^*$ is conjugate-linear (or linear from the conjugate space); the continuous Hilbert dual is introduced in Lecture 13.

Learning objectives

By the end of this lecture you will be able to:

- Construct the dual space V^* and its dual basis, and evaluate $\varphi(v) = \varphi_i v^i$.
- Use finite Riesz, including its complex conjugate-linearity, and distinguish it from the canonical linear map $V \to V^{**}$.
- Form the dual (transpose) map and compute with annihilators.
- Work with bilinear/multilinear forms and the covariant/contravariant index bookkeeping.

1 The dual space

Definition 11.2 (Dual space). A *linear functional* on V is a linear map $\varphi : V \to \mathbb{F}$. The set of all linear functionals, with pointwise addition and scalar multiplication, is the *dual space* $V^* = \mathcal{L}(V, \mathbb{F})$.

Example 11.3 (Functionals everywhere). On $\mathbb{F}^n$, $\varphi(x) = x_k$ extracts a coordinate. On $\mathcal{P}(\mathbb{F})$, $p \mapsto p(2)$ evaluates and $p \mapsto \int_0^1 p$ integrates. On $\mathcal{M}_n(\mathbb{F})$, $A \mapsto \operatorname{tr} A$. On an inner product space, $v \mapsto \langle u, v\rangle$ for fixed u. All are linear, so all live in V^*. ◁

Remark 11.4 (Invariant functionals). The trace is special among functionals on $\mathcal{M}_n$: it is invariant under similarity, $\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A$, so it descends to a functional on operators, not just matrices. Most functionals depend on the basis; the few that do not encode genuine invariants.

Proposition 11.5 (Dimension of the dual). If $\dim V = n$ then $\dim V^* = n$, so $V^* \cong V$ (non-canonically).

Proof. Fix a basis $e_1, \dots, e_n$ of V. A linear functional is determined by its values on the e_j, and those n values may be assigned independently, so $\varphi \mapsto (\varphi(e_1), \dots, \varphi(e_n))$ is an isomorphism $V^* \cong \mathbb{F}^n$. Hence $\dim V^* = n$. ■

The isomorphism with V is not canonical—it depends on a choice of basis—which is the whole point of treating V^* as a space in its own right.

Example 11.6 (Counting functionals). On $\mathbb{F}^3$ every functional is $\varphi(x) = a_1x_1 + a_2x_2 + a_3x_3$, fixed by the triple (a_1, a_2, a_3). Thus $V^* \cong \mathbb{F}^3$, matching $\dim V^* = \dim V = 3$—but the matching relies on the chosen coordinates and is not canonical. ◁

2 The dual basis

Definition 11.7 (Dual basis). Given a basis $e_1, \dots, e_n$ of V, the *dual basis* $e^1, \dots, e^n$ of V^* is defined by

$$e^i(e_j) = \delta^i_j = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

The functional e^i reads off the i-th coordinate: if $v = \sum_j v^j e_j$ then $e^i(v) = v^i$. The dual basis is a basis of V^*, and every functional expands as $\varphi = \sum_i \varphi(e_i)\, e^i$; writing $\varphi_i = \varphi(e_i)$,

$$\varphi(v) = \sum_i \varphi_i v^i.$$

The pairing of a covector's components φ_i with a vector's components v^i is the fundamental operation of the dual.

Example 11.8 (Computing a dual basis). Take $e_1 = (1, 1)$, $e_2 = (1, -1)$ in $\mathbb{F}^2$. Writing $e^1(x, y) = ax + by$, the conditions $e^1(e_1) = 1$, $e^1(e_2) = 0$ give $a + b = 1$, $a - b = 0$, so $e^1(x, y) = \frac{1}{2}(x + y)$; likewise $e^2(x, y) = \frac{1}{2}(x - y)$. The dual basis reads off the coordinates of (x, y) relative to e_1, e_2. ◁

Example 11.9 (Dual basis on polynomials). For $\mathcal{P}_2$ with basis $1, x, x^2$, the dual basis is $e^0(p) = p(0)$, $e^1(p) = p'(0)$, $e^2(p) = \frac{1}{2}p''(0)$: these return the Taylor coefficients, so $e^i(x^j) = \delta^i_j$. Evaluation, differentiation, and integration are all functionals expandable in this dual basis. ◁

Example 11.10 (Reconstructing a vector). The formula $v = \sum_i e^i(v)\, e_i$ rebuilds a vector from its coordinates. For $v = (3, 5)$ in the basis $e_1 = (1, 1)$, $e_2 = (1, -1)$ with the dual basis of Example 11.8, $e^1(v) = \frac{1}{2}(3 + 5) = 4$ and $e^2(v) = \frac{1}{2}(3 - 5) = -1$, so $v = 4e_1 - e_2$—and indeed $4(1, 1) - (1, -1) = (3, 5)$. ◁

Example 11.11 (Dual basis via matrix inverse). If the basis vectors are the columns of A, the dual-basis covectors are the *rows* of A^{-1}, because $(A^{-1}A)_{ij} = \delta_{ij}$ is exactly $e^i(e_j) = \delta^i_j$. For $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$,

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

so $e^1 = (1, -1)$ and $e^2 = (0, 1)$. The dual basis is the inverse-transpose construction in disguise. ◁

Remark 11.12 (Rows and columns). If vectors are columns, covectors are rows: $\varphi(v)$ is the row–column product $\varphi\, v$. If T has matrix A, a row covector pulls back as $\varphi \mapsto \varphi A$. If instead the coordinates of a covector are stored as a column, the same dual map is multiplication by A^{T}. Thus "transpose" depends only on the coordinate-storage convention; the invariant definition is $T'\varphi = \varphi \circ T$.

3 Functionals, the Riesz map, and the double dual

A nonzero functional φ has null φ of dimension $n-1$: its level sets are parallel hyperplanes, and $\varphi(v)$ counts (with sign and scale) how many it takes to reach v. This is the geometric picture of a covector.

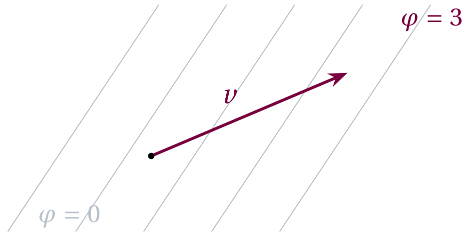

Figure 1: A linear functional as a family of parallel level hyperplanes ($\varphi = \text{const}$). The value $\varphi(v)$ measures how many sheets the vector v pierces—the "stack of planes" picture of a covector.

> **Proposition 11.13 (Finite Riesz comparison).** Let V be a finite-dimensional inner-product space over $\mathbb{F}$. Every algebraic functional $\varphi \in V^*$ has a unique representative $u \in V$ with $\varphi(v) = \langle u, v\rangle$ for all v. The map $R\colon u \mapsto \langle u, \cdot\rangle$ is linear over $\mathbb{R}$ and conjugate-linear over $\mathbb{C}$; equivalently, in the complex case it is a linear isomorphism $\overline{V} \to V^*$ from the conjugate vector space.

This is the finite theorem proved in Lecture 8. In Dirac notation $R(u) = \langle u|$, so a bra acts on $|v\rangle$ to give $\langle u \mid v\rangle$. It must not be confused with the canonical double-dual map below: R needs an inner product and is conjugate-linear over $\mathbb{C}$, whereas $J\colon V \to V^{**}$ needs no metric and is linear over both fields.

Example 11.14 (A functional as a vector). On $\mathbb{C}^2$ with the standard inner product, $\varphi(z_1, z_2) = 2z_1 - iz_2$ equals $\langle u, \cdot\rangle$ with $u = (2, i)$, since $\langle u, (z_1, z_2)\rangle = \overline{2}\,z_1 + \overline{i}\,z_2 = 2z_1 - iz_2$. The corresponding bra is the row vector $\langle u| = (2, -i)$. ◁

Example 11.15 (A functional and its hyperplane). The functional $\varphi(x, y, z) = 2x - y + 3z$ on $\mathbb{R}^3$ has kernel the plane $2x - y + 3z = 0$, a 2-dimensional hyperplane, with level sets $\varphi = c$ the parallel planes. The covector $(2, -1, 3)$ points along the normal, and $\varphi(v)$ grows with the signed distance from the kernel. ◁

Remark 11.16 (Algebraic versus continuous dual). The proposition is deliberately finite. On c_{00} with the ℓ^2 inner product, the algebraic functional $\varphi(x) = \sum_n x_n$ is well-defined but unbounded and cannot be $\langle u, \cdot\rangle$ for any finitely supported u. Lecture 13 states the Hilbert-space Riesz theorem for the *continuous* dual. In infinite-dimensional algebraic spaces V^* can be much larger than V, and the canonical map $V \to V^{**}$ below need not be onto.

> **Theorem 11.17 (The double dual).** For finite-dimensional V, the map $V \to V^{**}$ sending v to the functional $\varphi \mapsto \varphi(v)$ is a *canonical* isomorphism (no basis required).

Proof. Define $J\colon V \to V^{**}$ by $J(v)(\varphi) = \varphi(v)$; it uses no choice of basis. It is linear, since $J(av+bw)(\varphi) = \varphi(av+bw) = a\varphi(v)+b\varphi(w) = \big(aJ(v)+bJ(w)\big)(\varphi)$. It is injective: if $v \neq 0$, extend v to a basis $e_1 = v, e_2, \ldots, e_n$; the dual functional e^1 gives $J(v)(e^1) = e^1(v) = 1 \neq 0$, so $J(v) \neq 0$. Finally $\dim V^{**} = \dim V^* = \dim V$ by Proposition 11.5 applied twice, so the injective J between spaces of equal finite dimension is an isomorphism; as it used no basis, it is canonical. ■

So V and V^{**} may be identified outright: a vector is equally a thing functionals act on, and a thing that acts on functionals.

Example 11.18 (A vector acting on functionals). Read $v = (2, 3) \in \mathbb{F}^2$ as an element of V^{**}: it sends a functional $\varphi = \varphi_1 e^1 + \varphi_2 e^2$ to $\varphi(v) = 2\varphi_1 + 3\varphi_2$. Vector and double-dual act on each other by the same number, the symmetry $v(\varphi) = \varphi(v)$. ◁

Remark 11.19 (Bridge: the dual space in quantum mechanics). In the finite model, kets $|\psi\rangle$ live in V and bras $\langle\psi|$ live in V^*. In a Hilbert space a bra belongs to the continuous dual, not generally the full algebraic dual. A matrix element $\langle\phi| A |\psi\rangle$ pairs the bra $\langle\phi|$ (a functional) with the vector $A|\psi\rangle$. The whole bra–ket formalism says that physical amplitudes are functionals evaluated on states.

4 The dual map

Definition 11.20 (Dual map). For $T \in \mathcal{L}(V, W)$, the *dual map* $T' \in \mathcal{L}(W^*, V^*)$ is $T'\varphi = \varphi \circ T$, i.e. $(T'\varphi)(v) = \varphi(Tv)$.

The dual map pulls a measurement on W back to a measurement on V. It reverses arrows—$T: V \to W$ gives $T': W^* \to V^*$—and satisfies $(ST)' = T'S'$ and $(S + T)' = S' + T'$.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "T'\varphi:\, V \to \mathbb{F}"'] & W \arrow[d, "\varphi:\, W \to \mathbb{F}"] \\ \mathbb{F} \arrow[r, "\mathrm{id}_{\mathbb{F}}"'] & \mathbb{F} \end{tikzcd}$$

Figure 2: The typed evaluation square commutes: following $v \in V$ down gives $(T'\varphi)(v)$, while following it across and down gives $\varphi(Tv)$, so $(T'\varphi)(v) = \varphi(Tv)$.

Proposition 11.21 (The dual map is the transpose). In dual bases, $\mathcal{M}(T') = \mathcal{M}(T)^{\mathsf{T}}$: the matrix of the dual map is the transpose of the matrix of T.

Proof. Let $e_1, \ldots, e_n$ and $f_1, \ldots, f_m$ be the bases of V and W, with dual bases e^i and f^a, and write $Te_i = \sum_a M^a{}_i f_a$, so $M = \mathcal{M}(T)$. By Definition 11.20, the (i, a) entry of $\mathcal{M}(T')$ is

$$(T'f^a)(e_i) = f^a(Te_i) = f^a\Big(\textstyle\sum_b M^b{}_i f_b\Big) = M^a{}_i.$$

Thus the (i, a) entry of $\mathcal{M}(T')$ equals the (a, i) entry of $\mathcal{M}(T)$, i.e. $\mathcal{M}(T') = \mathcal{M}(T)^{\mathsf{T}}$. ■

This is the coordinate-free meaning of the transpose. The equality $\operatorname{rank} T' = \operatorname{rank} T$ is proved from annihilators below; that proof supplies the dual-space route to row rank equals column rank promised after Lecture 4's elementary RREF argument.

Example 11.22 (Dual map is the transpose). Let $T: \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$. For $\varphi = e^1$, $(T'e^1)(v) = e^1(Tv) = (Tv)_1 = v_1 + 2v_2$, so $T'e^1 = e^1 + 2e^2$; similarly $T'e^2 = 3e^2$. Hence $\mathcal{M}(T') = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 3 \end{smallmatrix}\right) = \mathcal{M}(T)^{\mathsf{T}}$. ◁

Example 11.23 (Transpose of a non-square map). A map $T\colon \mathbb{F}^3 \to \mathbb{F}^2$ with matrix $\left(\begin{smallmatrix}1&0&2\\0&1&3\end{smallmatrix}\right)$ has dual $T'\colon (\mathbb{F}^2)^* \to (\mathbb{F}^3)^*$ whose matrix is the 3×2 transpose

$$\mathcal{M}(T') = \begin{pmatrix}1&0\\0&1\\2&3\end{pmatrix},$$

sending a covector on $\mathbb{F}^2$ (a 2-component row) to one on $\mathbb{F}^3$. The dual swaps the roles of domain and codomain. ◁

Example 11.24 (Dual of the derivative). On $\mathcal{P}_2$ the derivative D has dual $D'\varphi = \varphi\circ D$. Taking $\varphi(p) = p(0)$, we get $(D'\varphi)(p) = \varphi(Dp) = (p')(0) = p'(0)$: the dual of differentiation sends "evaluate at 0" to "evaluate the derivative at 0." ◁

Example 11.25 (Dual map and the four subspaces). For $T\colon \mathbb{F}^3 \to \mathbb{F}^2$ of rank 2, range $T = \mathbb{F}^2$ gives $(\operatorname{range} T)^0 = \{0\} = \operatorname{null} T'$, so T' is injective; and $\operatorname{range} T' = (\operatorname{null} T)^0$ has dimension $3-1 = 2 = \operatorname{rank} T$. The dual map sees the same rank and complementary subspaces, with annihilators standing in for orthogonal complements. ◁

Example 11.26 (The dual reverses composition). For $S\colon U \to V$ and $T\colon V\to W$, $(TS)' = S'T'$: a functional on W is pulled back first along T, then along S. In matrices this is $(BA)^\mathsf{T} = A^\mathsf{T}B^\mathsf{T}$—the familiar order-reversal of the transpose, now seen as the order-reversal of pullback. ◁

Remark 11.27 (Dual map versus adjoint). The dual map T' needs no inner product; the adjoint $T^\dagger$ of Lecture 9 does. They are related by Riesz: identifying V^* with V via $u\mapsto \langle u,\cdot\rangle$ turns T' into $T^\dagger$, up to the conjugation built into the inner product. The transpose is the "bare" adjoint, before a metric is chosen.

Definition 11.28 (Annihilator). The *annihilator* of a subspace $U\subseteq V$ is $U^0 = \{\varphi\in V^* : \varphi(u) = 0 \text{ for all } u\in U\}$, a subspace of V^*. Algebraically $U^0 \cong (V/U)^*$; if V is finite-dimensional, this gives $\dim U + \dim U^0 = \dim V$.

Proposition 11.29 (Dual maps, annihilators, and rank). Let $T\colon V\to W$. Then $\operatorname{null} T' = (\operatorname{range} T)^0$ in arbitrary algebraic spaces. If V, W are finite-dimensional, then

$$\operatorname{rank} T' = \operatorname{rank} T, \qquad \operatorname{range} T' = (\operatorname{null} T)^0.$$

Consequently a finite matrix and its transpose have the same rank.

Proof. First, if $T'\varphi = 0$, then $\varphi(Tv) = 0$ for every v, so $\varphi\in(\operatorname{range} T)^0$; the converse is the same evaluation backwards. Also $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$, because for $u\in\operatorname{null} T$, $(T'\varphi)(u) = \varphi(Tu) = 0$. In finite dimensions the kernel identity and the annihilator dimension formula give

$$\operatorname{rank} T' = \dim W - \dim(\operatorname{range} T)^0 = \dim W - (\dim W - \operatorname{rank} T) = \operatorname{rank} T.$$

Rank–nullity also gives $\dim(\operatorname{null} T)^0 = \dim V - \dim\operatorname{null} T = \operatorname{rank} T$. The inclusion therefore joins two subspaces of the same finite dimension and is equality. In dual bases $\mathcal{M}(T') = \mathcal{M}(T)^\mathsf{T}$, so the rank equality is precisely row rank equals column rank. ■

These identities are the abstract, inner-product-free version of the four subspaces of Lecture 3.

Example 11.30 (An annihilator). For $U = \operatorname{span}\{e_1\} \subseteq \mathbb{F}^3$, the annihilator U^0 consists of all $\varphi = \varphi_1 e^1 + \varphi_2 e^2 + \varphi_3 e^3$ with $\varphi(e_1) = \varphi_1 = 0$; that is $U^0 = \operatorname{span}\{e^2, e^3\}$, of dimension $2 = 3 - 1$, confirming $\dim U + \dim U^0 = \dim V$. ◁

5 Bilinear and multilinear forms

Definition 11.31 (Bilinear form). A *bilinear form* is a map $B\colon V \times V \to \mathbb{F}$ linear in each argument. In a basis its matrix is $B_{ij} = B(e_i, e_j)$, and $B(u, v) = \sum_{i,j} u^i B_{ij} v^j = u^\mathsf{T} B v$.

A bilinear form is *symmetric* if $B(u, v) = B(v, u)$ (matrix $B^\mathsf{T} = B$) and *alternating* if $B(v, v) = 0$ (matrix antisymmetric). A symmetric bilinear form is essentially a (real) inner product when it is also positive definite; the quadratic form $Q(v) = B(v, v)$ recovers it by polarization.

Remark 11.32 (Bilinear versus sesquilinear). Over $\mathbb{C}$ the physics inner product is not bilinear but *sesquilinear*: conjugate-linear in the first slot (Lecture 7). A true bilinear form satisfies $B(\lambda u, v) = \lambda B(u, v)$ with no conjugate, so the inner product is a slightly different animal—and it is exactly this conjugation that makes the Riesz identification $V^* \cong V$ conjugate-linear over $\mathbb{C}$.

Definition 11.33 (Multilinear form). A k*-linear form* is a map $V \times \cdots \times V \to \mathbb{F}$ (k factors) linear in each slot separately. The determinant is the unique alternating n-linear form on $\mathbb{F}^n$ with $\det(e_1, \ldots, e_n) = 1$ (see the determinant facts recalled in Lecture 5; Example 11.36 below verifies the 2×2 case).

Example 11.34 (A bilinear form and its parts). The form $B(u, v) = u_1v_1 + 3u_1v_2 - u_2v_1$ has matrix $\left(\begin{smallmatrix} 1 & 3 \\ -1 & 0 \end{smallmatrix}\right)$, which splits into symmetric and antisymmetric parts

$$\begin{pmatrix} 1 & 3 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}.$$

The symmetric piece carries the quadratic-form content; over $\mathbb{R}^2$ the antisymmetric piece carries signed-area content. ◁

Example 11.35 (Evaluating a bilinear form). With $B = \left(\begin{smallmatrix} 2 & -1 \\ 0 & 3 \end{smallmatrix}\right)$, $u = (1, 2)$ and $v = (4, 1)$,

$$B(u, v) = u^\mathsf{T} B v = (1, 2) \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} = (1, 2) \begin{pmatrix} 7 \\ 3 \end{pmatrix} = 13.$$

A matrix sandwiched between a row and a column is a bilinear form in action. ◁

Example 11.36 (The determinant as a form). On $\mathbb{F}^2$ the determinant $D(u, v) = u_1v_2 - u_2v_1$ is bilinear, alternating ($D(v, v) = 0$), and normalized ($D(e_1, e_2) = 1$). It is the signed area of the parallelogram on u, v when $\mathbb{F} = \mathbb{R}$; algebraically over either course field, these three properties pin it down uniquely. ◁

Example 11.37 (Quadratic form and polarization). The symmetric form $B(u, v) = u_1v_1 + u_2v_2 + u_1v_2 + u_2v_1$ has quadratic form $Q(v) = B(v, v) = v_1^2 + v_2^2 + 2v_1v_2 = (v_1 + v_2)^2$. Polarization recovers it, $B(u, v) = \frac{1}{2}\big(Q(u + v) - Q(u) - Q(v)\big)$, so a symmetric bilinear form and its quadratic form carry the same information. ◁

Example 11.38 (Signature of a form). For a nondegenerate real symmetric form, the two-entry *signature* (p, q) records its positive and negative counts and is invariant under change of basis (Sylvester's law of inertia). A genuine inner product has signature $(n, 0)$; the Minkowski pseudo-metric below has signature $(1, 3)$. Degenerate forms require a third zero count. ◁

Example 11.39 (A volume form). On $\mathbb{R}^3$ the determinant $D(u, v, w) = \det[u\ v\ w]$ is an alternating 3-linear form giving the signed volume of the parallelepiped on u, v, w. Alternation encodes orientation: swapping two arguments flips the sign, and a repeated argument gives zero—a degenerate, volumeless box. ◁

Example 11.40 (An antisymmetric form). On $\mathbb{R}^2$ the form $\omega(u, v) = u_1 v_2 - u_2 v_1$ is alternating, with matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. It is the symplectic form of classical mechanics,

$$\omega\big((q, p), (q', p')\big) = qp' - pq',$$

pairing position with momentum; it underlies Hamilton's equations and the Poisson bracket. ◁

Example 11.41 (The Gram matrix of an inner product). For vectors $v_1, \ldots, v_k$ the Gram matrix $G_{ij} = \langle v_i, v_j \rangle$ is the matrix of the inner product restricted to their span. Over $\mathbb{R}$ the inner product is bilinear and G is symmetric; over $\mathbb{C}$ it is sesquilinear and G is Hermitian. In either case G is positive semidefinite, and $\det G \neq 0$ exactly when the v_i are linearly independent (Lecture 8). ◁

Remark 11.42 (Extension: covectors as differential forms). A smoothly varying covector field is a *differential* 1*-form*, such as df or the work integrand $F_i\, dx^i$. Integration pairs forms with curves, surfaces, and volumes, and the alternating multilinear forms of this lecture are the pointwise values of higher differential forms—the algebra beneath vector calculus and Stokes' theorem.

6 Covariant and contravariant components

Choose a basis and write a vector by its components v^i (upper index) and a covector by its components φ_i (lower index). Under a change of basis $e'_i = \sum_j P^j{}_i\, e_j$ the two transform *oppositely*: vector components by $[v]' = P^{-1}[v]$, while a fixed row covector has $[\varphi]' = [\varphi]P$ (equivalently its component column obeys $[\varphi]' = P^{\mathsf{T}}[\varphi]$). By contrast, the rows representing the *new dual-basis elements* in the old coordinates form P^{-1}. These are different questions: the basis elements and the components of one fixed covector must not be conflated. Components transforming opposite to the basis are called contravariant; covector components are covariant.

Remark 11.43 (Einstein summation). A repeated index appearing once up and once down is summed: $\varphi(v) = \varphi_i v^i$ means $\sum_i \varphi_i v^i$. The convention enforces a discipline—only an upper paired with a lower contracts to a scalar—that keeps expressions basis-independent and is the working language of relativity.

In a real inner-product space the positive-definite metric $g_{ij} = \langle e_i, e_j \rangle$ converts between the two: $\varphi_i = g_{ij} v^j$ lowers an index (turning a vector into the covector that is its inner product), and the inverse metric g^{ij} raises it. This is exactly the Riesz identification written in components, and it is why physics distinguishes the contravariant 4-vector x^μ from the covariant $x_\mu = g_{\mu\nu} x^\nu$. *Throughout this section the inner product is taken real* (the metric of tensor calculus and relativity), so $g_{ij} = g_{ji}$ and $\langle v, v \rangle = g_{ij} v^i v^j$ with no conjugation; over $\mathbb{C}$ the metric is Hermitian, $g_{ij} = \overline{g_{ji}}$, and index-lowering carries the conjugation of the conjugate-linear Riesz map of §3 (e.g. $\langle v, v \rangle = \overline{v^i}\, g_{ij} v^j$ and $g' = P^\dagger g P$). More generally, raising and lowering over $\mathbb{R}$ needs only a *nondegenerate symmetric pseudo-metric*; positivity is not required. The Minkowski form below is of this indefinite kind and is not an inner product.

Example 11.44 (Components transform oppositely). Rescale a basis vector to $e_1' = 2e_1$. A fixed vector $v = v^1 e_1 + \cdots$ then has new component $v'^1 = \frac{1}{2}v^1$ (contravariant—it shrinks as the basis grows), while the dual basis rescales as $e'^1 = \frac{1}{2}e^1$, so a covector component becomes $\varphi_1' = 2\varphi_1$ (covariant). Their product $\varphi_1 v^1$ is unchanged: the pairing is basis-independent. ◁

Example 11.45 (A non-diagonal invariant contraction). Let $E' = EP$ with $P = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$, and take $[v] = (2,3)^\mathsf{T}$ and the fixed row covector $[\varphi] = (4,-1)$. Then

$$[v]' = P^{-1}[v] = (-1,3)^\mathsf{T}, \qquad [\varphi]' = [\varphi]P = (4,3),$$

and $[\varphi]'[v]' = 4(-1) + 3(3) = 5 = [\varphi][v]$. The non-diagonal calculation exhibits the cancellation $PP^{-1} = I$ rather than merely asserting it. ◁

Example 11.46 (A mixed tensor transforms with both). A linear operator has components $T^i{}_j$ with one upper and one lower index, transforming under a change of basis as

$$T'^i{}_j = (P^{-1})^i{}_k \, T^k{}_l \, P^l{}_j,$$

which is the similarity $P^{-1}TP$ of Lecture 4, read as "raise the upper index by P^{-1}, lower the lower index by P." The matched mixed indices are exactly what make the trace $T^i{}_i$ invariant. ◁

Example 11.47 (A real metric is a $(0,2)$ tensor). The metric g_{ij} has two lower indices, so it is a $(0,2)$ tensor—a bilinear form. Lowering, $v_i = g_{ij}v^j$, contracts a contravariant vector against one slot of g to produce a covector. The metric is the universal device for trading a vector for the covector it represents. ◁

Example 11.48 (Bridge: raising and lowering with a pseudo-metric). With the Minkowski metric $g_{\mu\nu} = \operatorname{diag}(1,-1,-1,-1)$, a contravariant 4-vector $x^\mu = (t,x,y,z)$ has covariant form

$$x_\mu = g_{\mu\nu}x^\nu = (t,-x,-y,-z), \qquad x_\mu x^\mu = t^2 - x^2 - y^2 - z^2,$$

the last being the invariant interval. This nondegenerate symmetric form has signature $(1,3)$ but is indefinite, so it is a pseudo-metric rather than a positive inner product. Its inverse $g^{\mu\nu}$ raises indices. ◁

Example 11.49 (Bridge: the differential and gradient). The differential df eats a vector v and returns the directional derivative $df(v) = \sum_i \partial_i f\, v^i$, so its components $\partial_i f$ carry a *lower* index: a gradient is naturally a covector. Promoting it to "the gradient vector" secretly uses the metric to raise the index, which is why the steepest-ascent direction depends on the inner product. ◁

Example 11.50 (A metric that is not the identity). In the basis $e_1 = (1,0)$, $e_2 = (1,1)$ of $\mathbb{R}^2$ the metric components are

$$g_{ij} = \langle e_i, e_j\rangle = \begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}.$$

Covariant components are $v_i = g_{ij}v^j$, and the inner product is $\langle v,v\rangle = g_{ij}v^iv^j$—read off the metric, not assumed to be a sum of squares. ◁

Example 11.51 (Raising with the inverse metric). The metric $g = \left(\begin{smallmatrix}1 & 1\\ 1 & 2\end{smallmatrix}\right)$ above has inverse $g^{-1} = \left(\begin{smallmatrix}2 & -1\\ -1 & 1\end{smallmatrix}\right)$, so a covector raises to a vector by $\varphi^i = g^{ij}\varphi_j$; e.g. $(1,0) \mapsto (2,-1)$. Lowering then raising returns the original components, since $g^{ij}g_{jk} = \delta^i_k$. ◁

Example 11.52 (Inner product through the metric). With the metric $g = \left(\begin{smallmatrix}1 & 1\\ 1 & 2\end{smallmatrix}\right)$ above and basis vectors $u = e_1$, $v = e_2$ (contravariant components $(1,0)$ and $(0,1)$),

$$\langle u, v\rangle = g_{ij}u^iv^j = g_{12} = 1,$$

so e_1 and e_2 are *not* orthogonal—a fact the metric records but the bare component lists conceal. ◁

Example 11.53 (Bridge: reciprocal basis). In an inner product space the dual basis can itself be drawn as vectors: the *reciprocal basis* of $a_1, \ldots, a_n$ is the unique set $b^1, \ldots, b^n$ with $\langle b^i, a_j \rangle = \delta^i_j$—the Riesz vectors of the dual basis, computed by raising an index, $b^i = g^{ij} a_j$. For the oblique pair $a_1 = (1,0)$, $a_2 = (1,1)$ of Example 11.50, the inverse metric of Example 11.51 gives

$$b^1 = 2a_1 - a_2 = (1, -1), \qquad b^2 = -a_1 + a_2 = (0, 1),$$

and indeed $\langle b^i, a_j \rangle = \delta^i_j$. Each b^i is orthogonal to every a_j with $j \neq i$ and normalized against a_i. In crystallography the same construction (with a conventional factor 2π, $\mathbf{b}^i \cdot \mathbf{a}_j = 2\pi \delta^i_j$) produces the *reciprocal lattice* of a crystal, the natural home of wavevectors and of X-ray diffraction peaks—a covector construction every solid-state physicist uses daily. ◁

Remark 11.54 (Bridge: momentum is a covector). In mechanics the momentum $p_i = \partial L / \partial \dot{q}^i$ carries a lower index: it is dual to the velocity $\dot{q}^i$, and the pairing $p_i \dot{q}^i$ is the invariant Legendre scalar with energy units in $H = p_i \dot{q}^i - L$. Generalized force paired with velocity, $Q_i \dot{q}^i$, is power. Wavevectors, gradients, and momenta are covectors; positions and velocities are vectors. The distinction, invisible in an orthonormal Cartesian frame, becomes essential in curved or relativistic settings.

Remark 11.55 (Towards tensors). A vector carries one upper index, a covector one lower, a bilinear form two lower, a linear operator one of each. All are *tensors*, classified by how many indices of each type they carry and how those indices transform. Lecture 12 builds them systematically from tensor products.

Summary of main concepts

The dual space $V^* = \mathcal{L}(V, \mathbb{F})$ collects the linear functionals on V; a basis e_i of V gives a dual basis e^i with $e^i(e_j) = \delta^i_j$, and $\varphi(v) = \varphi_i v^i$. In a finite inner-product space Riesz identifies V^* with V conjugate-linearly over $\mathbb{C}$ (bras are dual vectors), while the double-dual map $V \to V^{**}$ is canonical and linear. A map $T\colon V \to W$ has a dual $T'\colon W^* \to V^*$, $T'\varphi = \varphi \circ T$, whose matrix is the transpose of T's, with null $T' = (\operatorname{range} T)^0$. Bilinear and multilinear forms package several vector inputs into a scalar—the determinant being the alternating n-linear form. Vector components v^i are contravariant, covector components φ_i covariant, summed by Einstein's convention and related through the metric g_{ij}—the bookkeeping that tensors (Lecture 12) formalize.

- **Dual space** $V^* = \mathcal{L}(V, \mathbb{F})$—the linear functionals on V; $\dim V^* = \dim V$.
- **Dual basis**—$e^i(e_j) = \delta^i_j$; $\varphi(v) = \varphi_i v^i$.
- **Riesz map and double dual**—finite complex Riesz is conjugate-linear; $V \to V^{**}$ is canonical and linear.
- **Dual map**—$T'\varphi = \varphi \circ T$; its matrix is the transpose; null $T' = (\operatorname{range} T)^0$.
- **Bilinear and multilinear forms**—several vector inputs to a scalar; the determinant is the alternating n-linear form.
- **Covariant vs contravariant**—components v^i (contra) and φ_i (co), summed by Einstein's convention and related by the metric g_{ij}.

Exercises for this lecture are in the companion sheet `exercises-11.pdf`.